

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
reasoning process here
wavelength = $h / p = h / \sqrt{2 m \hat{\mu}}$
using the relation $E = \frac{1}{2} m v^2 = \frac{1}{2} m (h / (\text{wavelength}))^2 = \hat{\mu}$, we can find v^2
hence, $\sqrt{2 \hat{\mu} / m} = v$, plugging into wavelength equation
</think>
<answer>
G
</answer>

